The problem

One pump change moves everything.

- Raise a pump's speed and the outlet valve, tank, steam coil and shared steam header all respond.
- Each system watches only its own piece.
- The lesson from the last incident sits in a log nobody reads at change time.

Feed pump	**Hold tank**	**Outlet valve**	**Steam header**
more flow	level and heat load shift	opens toward its limit	shared with the other plant

The product

Negotiate. Remember. Refuse.

- **Negotiate:** every setpoint change is a proposal, and every connected machine's agent votes.
- **Remember:** each agent cites past runs, then is scored against what really happened.
- **Refuse:** one no blocks the change with evidence. A safer target follows, and a human gets a 10-second veto.
- **Across plants:** a rule from Plant A is cited in Plant B.

Pump 50% to 85%
Blocked: the outlet valve would reach 89%, past its 81% line

Counter: 50% to 65%
5 of 5 yes, veto window, one write to the simulator

Plant B at 85%
Blocked, citing the rule from Plant A's runs

Why both tools

Actors hold the truth. Relay carries the argument.

Durable Actors (Terse)

- One actor per component, plant and company
- State, history, calibration and rules live in the actor
- One call at a time per actor, so decisions stay in order
- A setpoint needs a one-time token from the execute path

Agent Relay

- Ten component agents on plant and company channels
- Proposals, votes and counters travel as messages
- Cross-plant votes go over the company channel
- Results reach the agents, which score their predictions

Shown on the full stack: 26 of 26 end-to-end checks on real actors, Relay and model calls, on the final code.